CurrentMeasuerment_1
Measurement.SchDoc

I_OUT_N — I1_OUT_N
I_OUT_P — I1_OUT_P

CurrentMeasuerment_2
Measurement.SchDoc

I_OUT_N — I2_OUT_N
I_OUT_P — I2_OUT_P

CurrentMeasuerment_3
Measurement.SchDoc

I_OUT_N — I3_OUT_N
I_OUT_P — I3_OUT_P

CurrentMeasuerment_4
Measurement.SchDoc

I_OUT_N — I4_OUT_N
I_OUT_P — I4_OUT_P

VIN

C45 2.2 µF 100 VDC
C46 2.2 µF 100 VDC
C47 2.2 µF 100 VDC
C48 2.2 µF 100 VDC
C49 2.2 µF 100 VDC

GND

VIN

C14 2.2 µF 100 VDC
C34 2.2 µF 100 VDC
C41 2.2 µF 100 VDC
C42 2.2 µF 100 VDC
C43 2.2 µF 100 VDC

GND

GND GND

R19 1M
C54 100pF

J5
S2
S1

I1_OUT_N A8
I1_OUT_P A7
I2_OUT_N A5
I2_OUT_P A4
I3_OUT_N A6
I3_OUT_P A3
I4_OUT_N A2
I4_OUT_P A1

R-RJ45R10P-B000

Power_Supply
Versorgungsspannung

U_Spannungswandlung
Spannungswandlung.SchDoc

MH5 3.4mm Mounting Hole
MH6 3.4mm Mounting Hole
MH1 3.4mm Mounting Hole
MH2 3.4mm Mounting Hole
MH4 3.4mm Mounting Hole

LOGO1
UltraZohm
UZ Logo

INFO1
Project
ProjectRevision
AuthorParam
ProjectDate
Design Information

INFO2
ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv
ZC Serialnumber

Title TopSheet.SchDoc		UltraZohm www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_hc5fw.PrjPcb		Date: 14/10/2026 Sheet 1 of 7

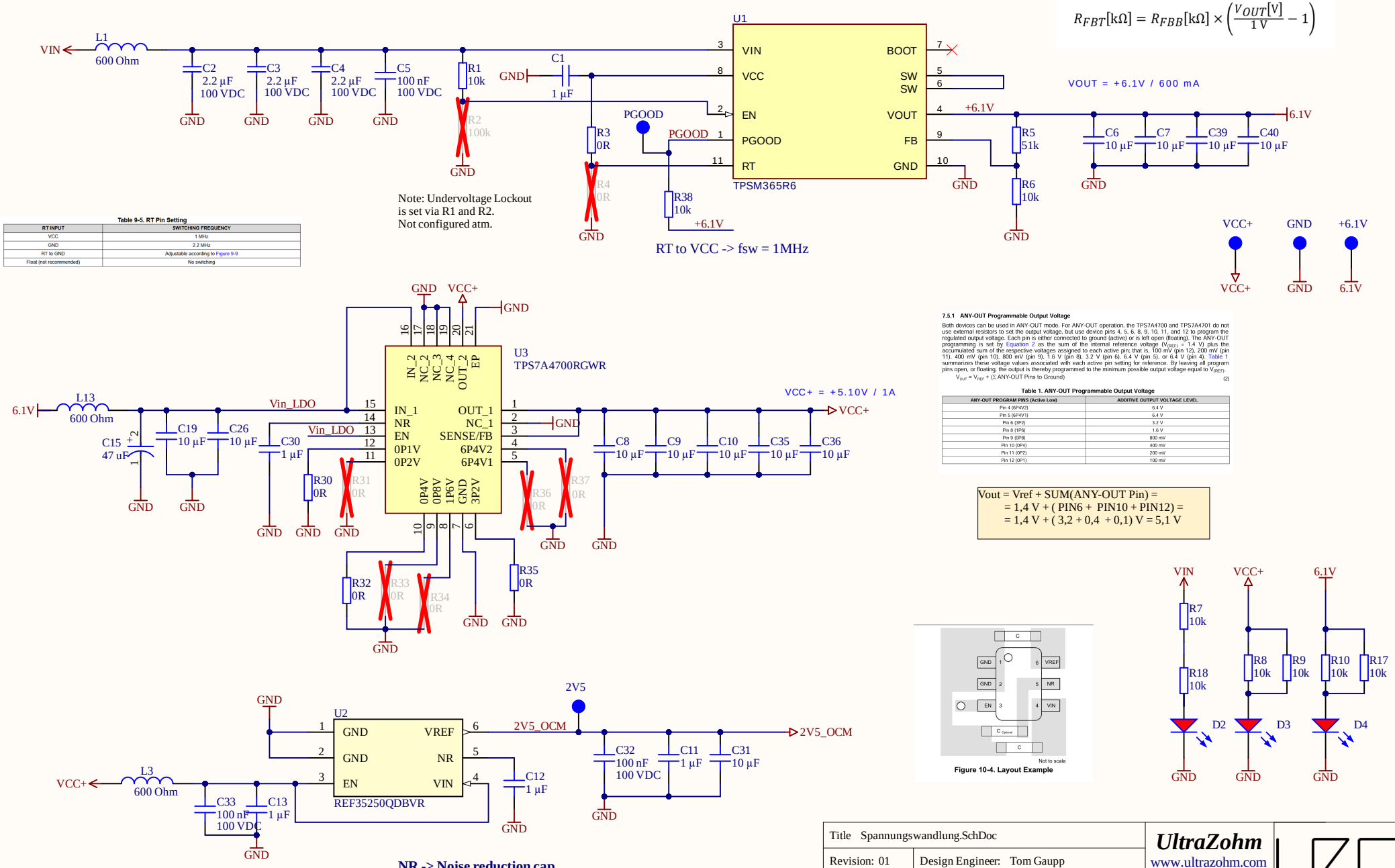


Table 9-5. RT Pin Setting

RT INPUT	SWITCHING FREQUENCY
VCC	1 MHz
GND	2.2 MHz
RT to GND	Adjustable according to Figure 9-9
Float (not recommended)	No switching

7.5.1 ANY-OUT Programmable Output Voltage

Both devices can be used in ANY-OUT mode. For ANY-OUT operation, the TPS7A4700 and TPS7A4701 do not use external resistors to set the output voltage, but use device pins 4, 5, 6, 8, 9, 10, 11, and 12 to program the regulated output voltage. Each pin is either connected to ground (active) or is left open (floating). The ANY-OUT programming is set by Equation 2 as the sum of the internal reference voltage (V_{REF}) = 1.4 V) plus the accumulated sum of the respective voltages assigned to each active pin; that is, 100 mV (pin 12), 200 mV (pin 11), 400 mV (pin 10), 800 mV (pin 9), 1.6 V (pin 8), 3.2 V (pin 6), 6.4 V (pin 5), or 6.4 V (pin 4). Table 1 summarizes these voltage values associated with each active pin setting for reference. By leaving all program pins open, or floating, the output is thereby programmed to the minimum possible output voltage equal to V_{REF}.

$V_{OUT} = V_{REF} + (\Sigma \text{ ANY-OUT Pins to Ground})$ (2)

Table 1. ANY-OUT Programmable Output Voltage

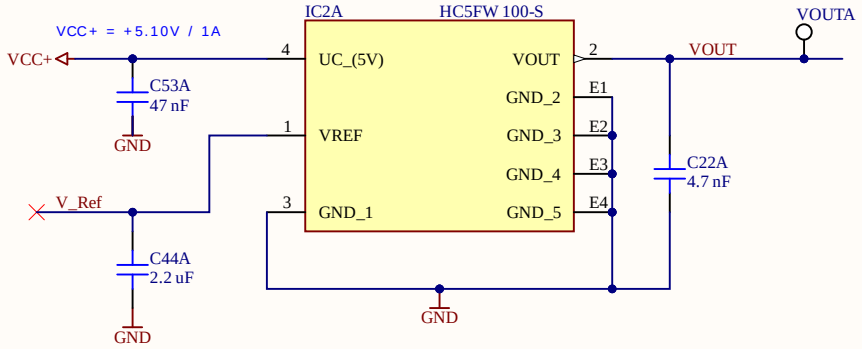
ANY-OUT PROGRAM PINS (Active Low)	ADDITIVE OUTPUT VOLTAGE LEVEL
Pin 4 (6P4V2)	6.4 V
Pin 5 (6P4V1)	6.4 V
Pin 6 (3P2)	3.2 V
Pin 8 (1P6)	1.6 V
Pin 9 (0P8)	800 mV
Pin 10 (0P4)	400 mV
Pin 11 (0P2)	200 mV
Pin 12 (0P1)	100 mV

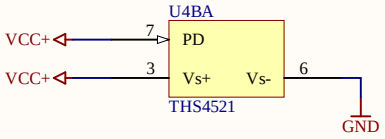
Vout = Vref + SUM(ANY-OUT Pin) =
= 1,4 V + (PIN6 + PIN10 + PIN12) =
= 1,4 V + (3,2 + 0,4 + 0,1) V = 5,1 V

Figure 10-4. Layout Example

NR -> Noise reduction cap

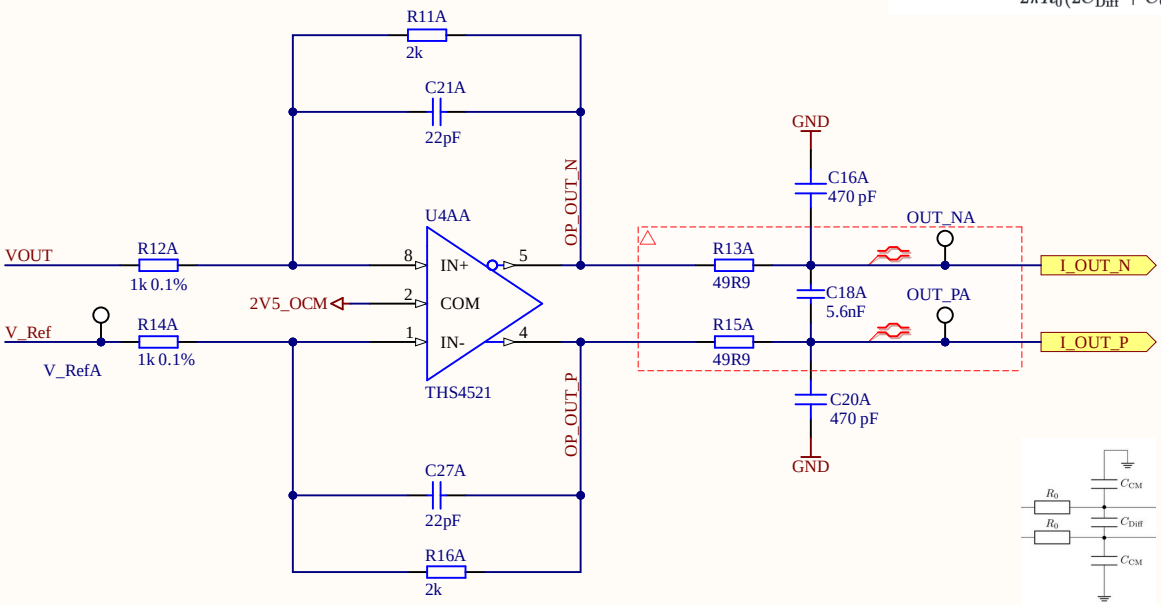
Title Spannungswandlung.SchDoc		UltraZohm www.ultrazohm.com
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Project: uz_per_current_measurement_hc5fw.PrjPcb		Date: 14/10/2026 Sheet 2 of 7

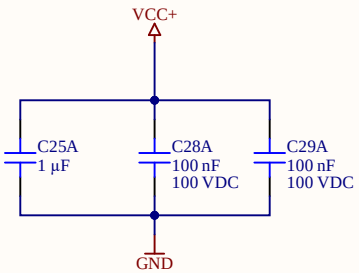


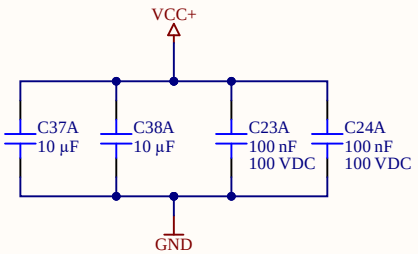


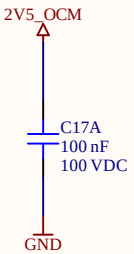
$$f_{\text{aliasing},-3\text{dB}} = \frac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$$

f_cut,off = 273 kHz



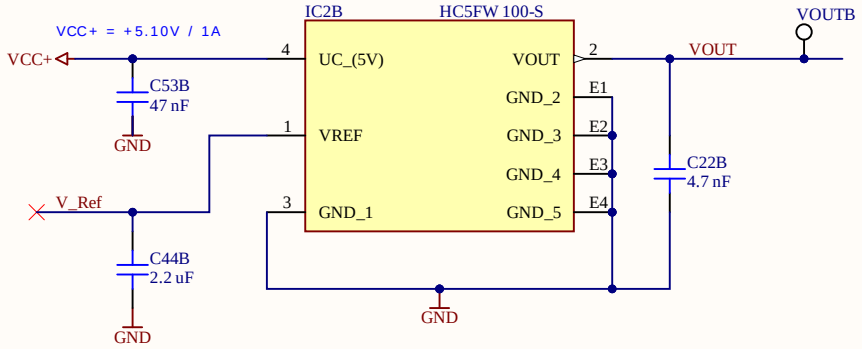


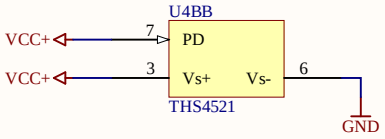






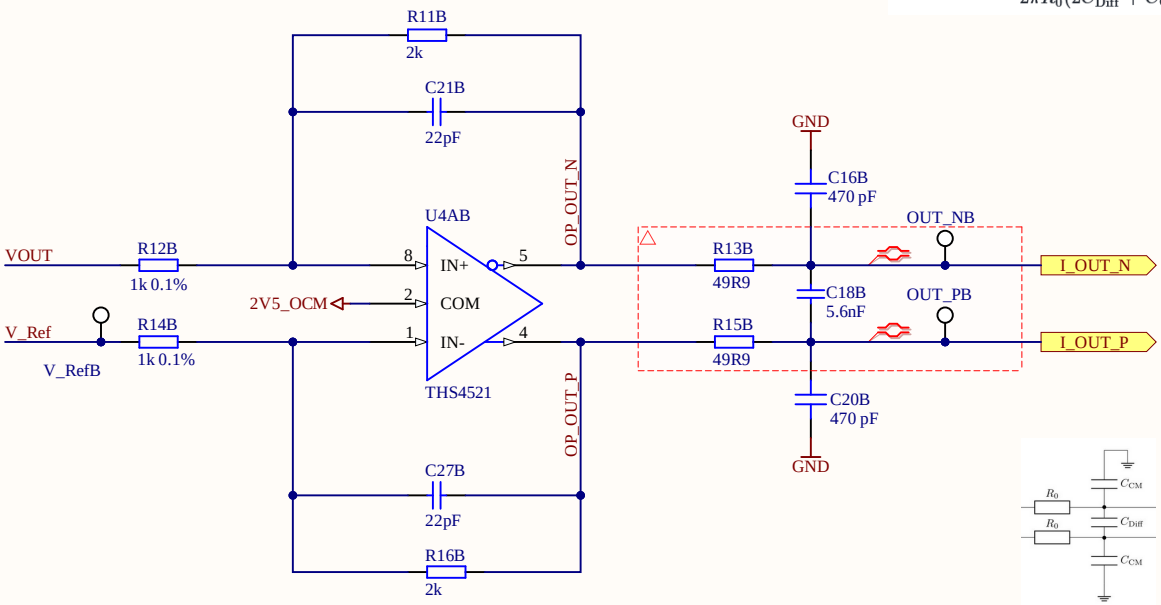
Title Measurement.SchDoc		UltraZohm	
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Project: uz_per_current_measurement_hc5fw.PrjPcb		Date: 14/10/2026	
		Sheet 3.1 of 7	

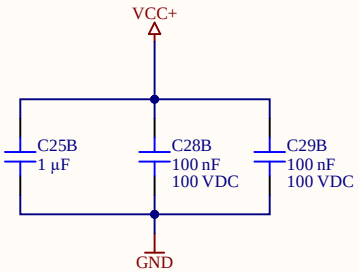


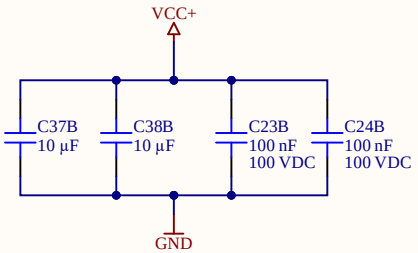


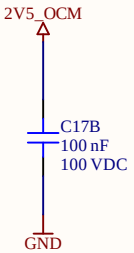
$$f_{aliasing,-3dB} = \frac{1}{2\pi R_0 (2C_{Diff} + C_{CM})}$$

f_cut,off = 273 kHz



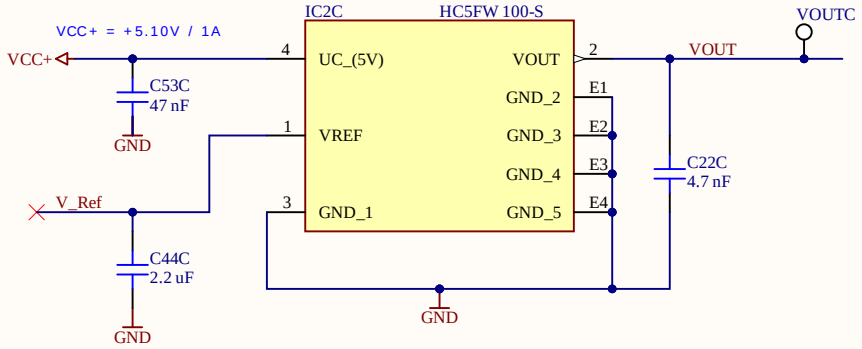


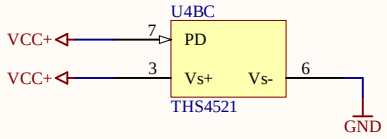






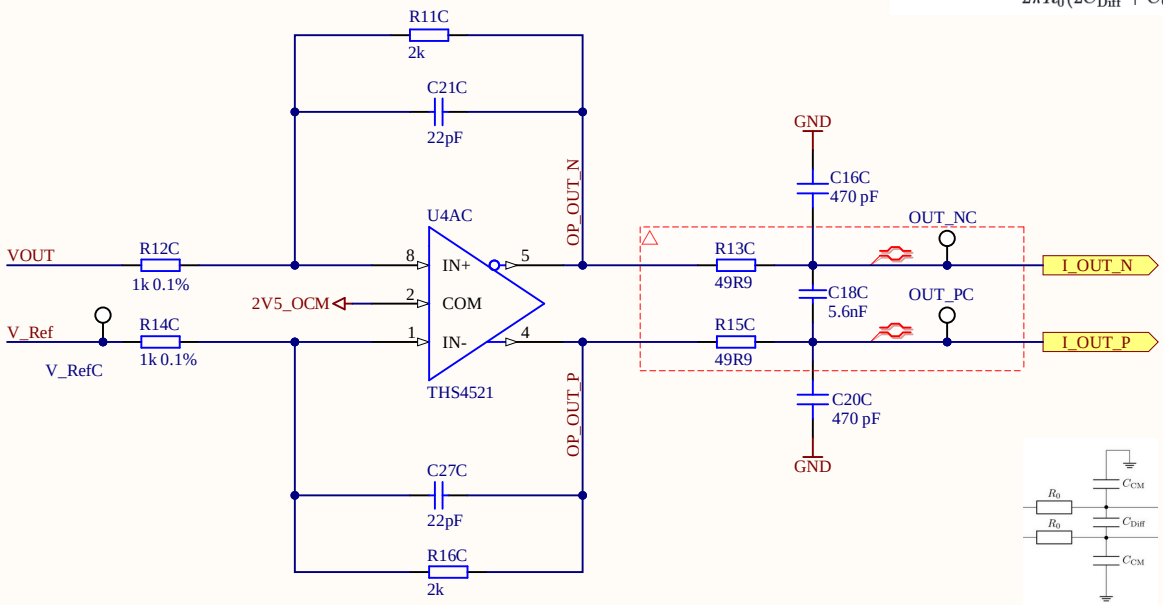
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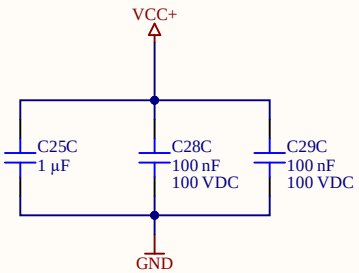


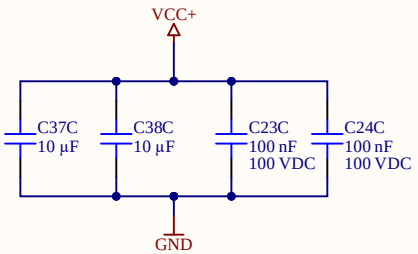


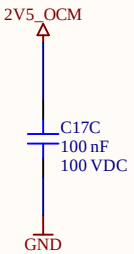
$$f_{\text{aliasing},-3\text{dB}} = \frac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$$

f_cut,off = 273 kHz



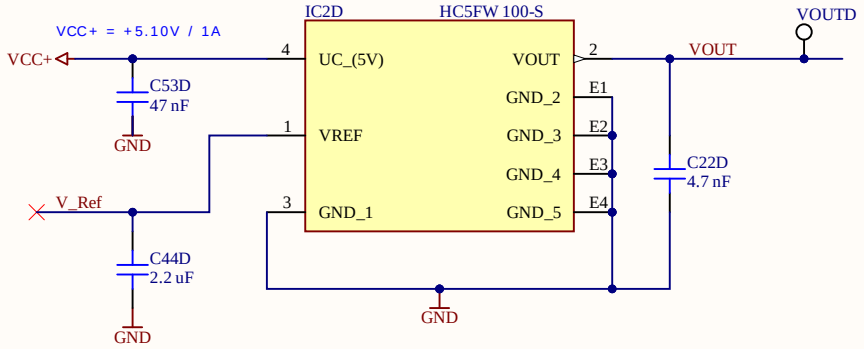


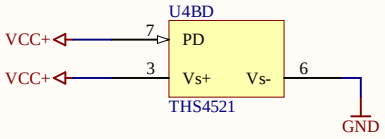






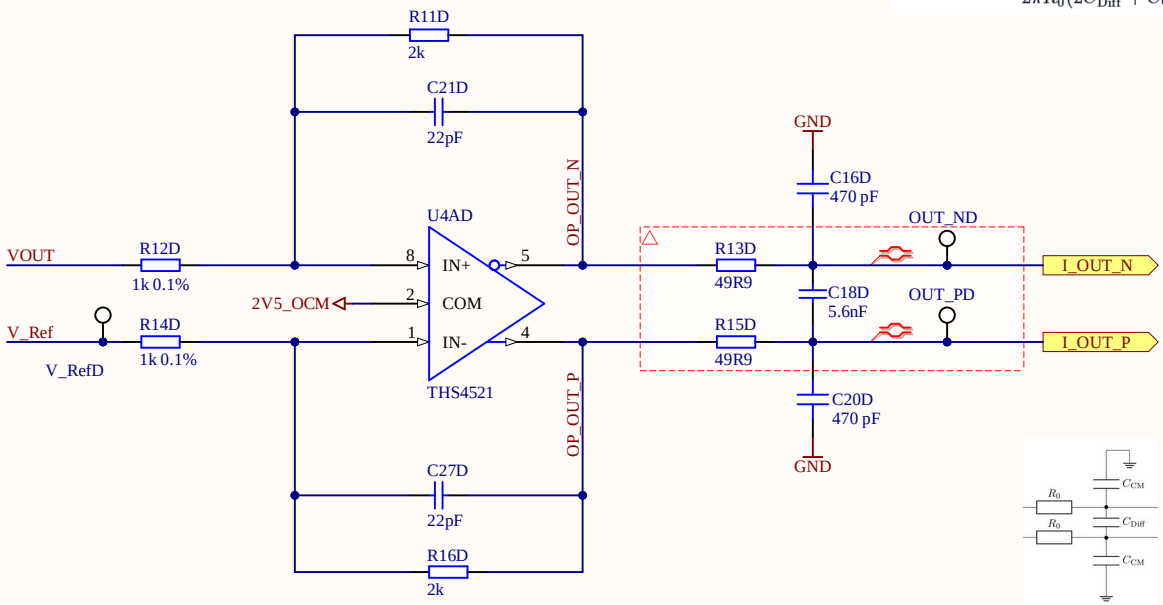
Title Measurement.SchDoc		UltraZohm	
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Project: uz_per_current_measurement_hc5fw.PrjPcb		Date: 14/10/2026	
		Sheet 3.3 of 7	

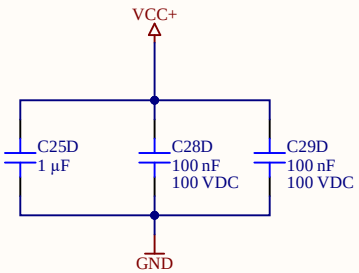


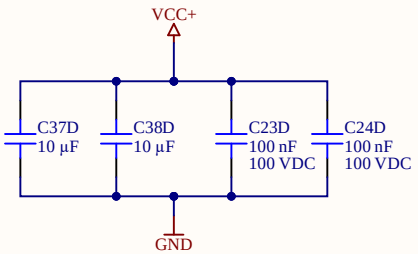


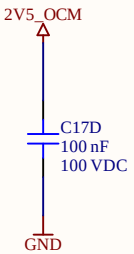
$$f_{\text{aliasing},-3\text{dB}} = \frac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$$

f_cut,off = 273 kHz



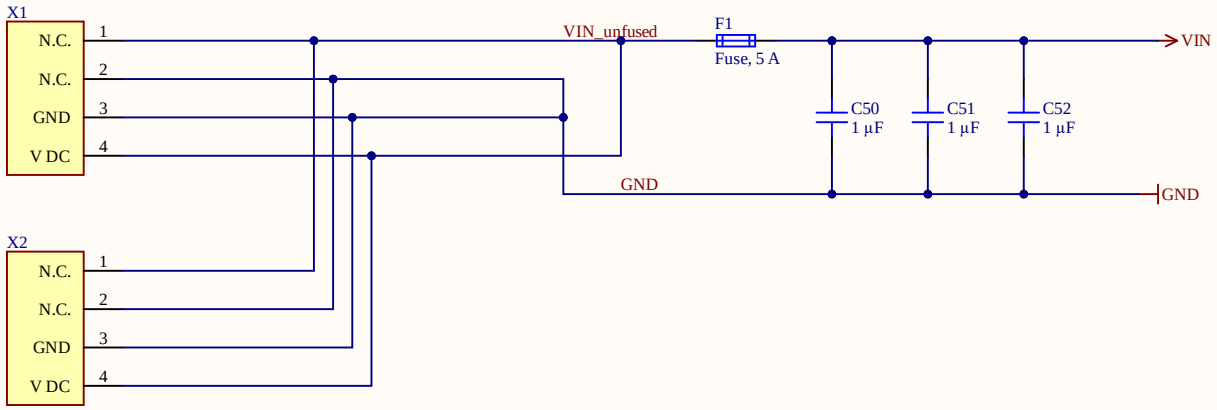








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@X1 & @X2 (M8 Buchsen für Stromversorgung):

Pin 1 auf Pin 4 gebrückt und
Pin 2 auf Pin 3 gebrückt

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